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United States Patent

12404679

Kind Code

B2

Date of Patent

September 02, 2025

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Building covering system incorporating a building panel, and methods of forming and installing the same

Abstract

Described herein is a building system comprising: a plurality of support members each comprising a support surface, a plurality of first building panels secured to the support surface of the plurality of support members, each of the plurality of first building panels comprising a non-porous body, a plurality of second building panels secured to the plurality of first building panels by an adhesive and a first plurality of mechanical fasteners, each of the plurality of the second building panels comprising a porous body.

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Appl. No.: 17/894562

Filed: August 24, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230060772 A1	Mar. 02, 2023

Related U.S. Application Data

us-provisional-application US 63237325 20210826

Publication Classification

Int. Cl.: E04B2/00 (20060101); E04F13/08 (20060101)

U.S. Cl.:

CPC E04F13/0866 (20130101);

Field of Classification Search

CPC: E04F (13/0866); E04F (13/0803)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2126956	12/1937	Gilbert	N/A	N/A
3084402	12/1962	Jordan	N/A	N/A
3339784	12/1966	Filstead, Jr.	220/560.11	F17C 3/025
3879240	12/1974	Wall	264/46.7	B60P 3/32
6012255	12/1999	Smid	52/745.19	E04C 2/043
8925677	12/2014	Dugan	N/A	N/A
10221562	12/2018	Fay	N/A	N/A
2003/0000901	12/2002	Salatin	211/85.16	A47F 5/0846
2005/0055935	12/2004	Layfield	52/506.01	E04B 2/7411
2006/0000670	12/2005	Dodd	N/A	N/A
2006/0019568	12/2005	Toas	N/A	N/A
2006/0191743	12/2005	Pike	N/A	N/A
2012/0291384	12/2011	Martens	N/A	N/A
2014/0162224	12/2013	Wallace	434/219	G09B 5/06
2015/0275518	12/2014	Flick	428/189	E04C 2/26
2017/0037639	12/2016	Taylor	N/A	N/A
2020/0002938	12/2019	Buck	N/A	N/A
2020/0173172	12/2019	Kragness	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2840195	12/2014	EP	N/A
2018/0025688	12/2017	KR	N/A
10-2020-0046936	12/2019	KR	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 63/237,325, filed Aug. 26, 2021, the entirety of which is incorporated herein by reference.

BACKGROUND

(1) Creating an aesthetically pleasing and structurally stable building system may require both

skilled labor as well as specialty building hardware that may drive up the cost of installation. Therefore, a need exists for a building system that may exhibit the desired aesthetic characteristics that may be installed without necessitating specialty hardware and/or specialized labor to create the finished system.

BRIEF SUMMARY

(2) The present invention is directed to a building system comprising: a plurality of support members each comprising a support surface; a plurality of first building panels secured to the support surface of the plurality of support members, each of the plurality of first building panels comprising a non-porous body; a plurality of second building panels secured to the plurality of first building panels by an adhesive and a plurality of first mechanical fasteners, each of the plurality of the second building panels comprising a porous body.

(3) Other embodiments of the present invention include a building system comprising: a plurality of support members; a plurality of first building panels each having a first major surface opposite a second major surface and a side surface extending there-between; a plurality of second building panels each having a first major surface opposite a second major surface and a side surface extending there-between; wherein the plurality of first building panels are secured to the plurality of support members; wherein the plurality of second building panels are secured to the plurality of first building panels by an adhesive; and wherein the plurality of second building panels and the plurality of first building panels arranged in an offset pattern such that the side surface of at least one of the plurality of first building panels and the side surface of at least one of the plurality of second building panels are not co-planar.

(4) Other embodiments of the present invention include a method of installing a building system comprising: a) bringing into contact a second major surface of at least one facing building panel and a pre-existing building surface of a pre-existing building layer to form a building interface, whereby adhesive is present in the interface; and b) driving at least one mechanical fastener into the at least one facing building panel such that the at least one mechanical fastener is located at least in a portion of the at least one facing building panel and the pre-existing building surface; wherein the at least one building panel comprises a porous body.

(5) Other embodiments of the present invention include a method of installing a building system comprising: a) bringing into contact a plurality of facing building panels with a pre-existing building layer to form a building interface, whereby adhesive is present in the interface; and b) driving at least one mechanical fastener into the at least one facing building panel such that the at least one mechanical fastener is located at least in a portion of the at least one building panel and the pre-existing building surface; wherein the pre-existing building layer is formed of a plurality of backing building panels located immediately adjacent to each other; and wherein the plurality of facing building panels are arranged in an offset pattern relative to the plurality of backing building panels such that at least one edge of the facing building panels and at least one edge of the backing building panels are not co-planar.

(6) Other embodiments of the present invention include a building system comprising: a plurality of support members comprising a support surface; a plurality of building panels each having a first major surface opposite a second major surface and a side surface extending there-between; wherein the plurality of building panels are secured to the plurality of support members by an adhesive and a plurality of fasteners; and wherein each of the plurality of building panels comprises a porous body, the porous body comprising a fibrous material.

(7) Other embodiments of the present invention include a building system comprising: a plurality of support members comprising a support surface; a plurality of building panels each having a first major surface opposite a second major surface and a side surface extending there-between, each of the plurality of building panels having a panel thickness as measured from the first major surface to the second major surface of each of the plurality of building panels; wherein the plurality of building panels are secured to the plurality of support members by an adhesive and a plurality of

fasteners; and wherein each of the plurality of fasteners has a fastener length as measured from a proximal end of the fastener to a distal end of the fastener and each of the plurality of fasteners has a head having an outer diameter; and wherein the distal end of each of the plurality of fasteners is located in the support member and the proximal end of each of the plurality of fasteners is located inset of the first major surface of the building panel by a depth that is a non-zero value.

(8) Other embodiments of the present invention include a method of installing a building system comprising: a) bringing into contact a plurality of building panels with a pre-installed support grid comprising a support surface to form a first interface there-between, whereby adhesive is present in the first interface and each of the plurality of building panels has a panel thickness; and b) driving at least one mechanical fastener into the at least one building panel such that a distal end of the mechanical faster is inset of the first major surface by a depth that is a non-zero value, whereby the mechanical fastener extends through the first interface;

(9) Further areas of applicability of the present invention will become apparent from the detailed description provided hereinafter. It should be understood that the detailed description and specific examples, while indicating the preferred embodiment of the invention, are intended for purposes of illustration only and are not intended to limit the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present invention will become more fully understood from the detailed description and the accompanying drawings, wherein:
- (2) FIG. 1 is top perspective view of a first building panel according to the present invention;
- (3) FIG. 2A is top perspective view of a second building panel according to the present invention;
- (4) FIG. 2B is top view of the second building panel of FIG. 2A;
- (5) FIG. 3A is a building system according to one embodiment of the present invention that comprises the first and second building panels of the present invention;
- (6) FIG. 3B is a building system according to another embodiment of the present invention that comprises the second building panels of the present invention;
- (7) FIG. 4A is a close-up view of region IV shown in FIG. 3A;
- (8) FIG. 4B is a close-up view of region V shown in FIG. 3B;
- (9) FIG. 5A is an upward facing perspective view of the building system of FIG. 3A during installation;
- (10) FIG. 5B is an upward facing perspective view of the building system of FIG. 3B during installation;
- (11) FIG. 6 is an upward facing perspective view of the building system during installation;
- (12) FIG. 7 is an upward view of the first building panel with support members installed into the building system of FIG. 3 being installed;
- (13) FIG. 8A is an upward view of the second building panel with first building panel of the building system of FIG. 3A being installed according to one embodiment;
- (14) FIG. 8B is an upward view of the second building panel with first building panel of the building system of FIG. 3A being installed according to another embodiment;
- (15) FIG. 8C is an upward view of the second building panel with first building panel of the building system of FIG. 3B being installed according to another embodiment;
- (16) FIG. 8D is an upward view of the second building panel with first building panel of the building system of FIG. 3B being installed according to one embodiment;
- (17) FIG. 9 is an upward facing perspective view of the building system during installation;
- (18) FIG. 10 is an upward facing perspective view of the building system during installation; and
- (19) FIG. 11 is an upward facing perspective view of the building system during installation.

DETAILED DESCRIPTION

(20) The following description of the preferred embodiment(s) is merely exemplary in nature and is in no way intended to limit the invention, its application, or uses.

(21) As used throughout, ranges are used as shorthand for describing each and every value that is within the range. Any value within the range can be selected as the terminus of the range. In addition, all references cited herein are hereby incorporated by referenced in their entireties. In the event of a conflict in a definition in the present disclosure and that of a cited reference, the present disclosure controls.

(22) The description of illustrative embodiments according to principles of the present invention is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the invention disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top,” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such.

(23) Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. Moreover, the features and benefits of the invention are illustrated by reference to the exemplified embodiments. Accordingly, the invention expressly should not be limited to such exemplary embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features; the scope of the invention being defined by the claims appended hereto.

(24) Unless otherwise specified, all percentages and amounts expressed herein and elsewhere in the specification should be understood to refer to percentages by weight. The amounts given are based on the active weight of the material. According to the present application, the term “about” means $\pm 5\%$ of the reference value. According to the present application, the term “substantially free” less than about 0.1 wt. % based on the total of the referenced value.

(25) Referring to FIGS. 3A and 4A, the building system **1** of the present invention may comprise a plurality of first building panels **100**, a plurality of second building panels **200**, and one or more support elements **50** (also referred to as a “support member”). The support element **50** may comprise a support surface **51**.

(26) Non-limiting examples of support structures include suspended ceiling grids, wall-studs (also called framing studs), and structural walls (e.g., ceiling formed by an adjacent subfloor in a multi-floor building, roof). In the non-limiting exemplified embodiment, the support structure **50** is a ceiling joist—whereby the ceiling joist forms the support surface **51**. Although not shown, in other embodiments, the support element **50** may comprise an inverted T-bar having a horizontal flange and a vertical web—whereby the horizontal flange may form the support surface **51**.

(27) The building system **1** may be installed in an interior space, whereby the interior space comprises a plenary space **3** and an active room environment **2**. The plenary space **3** provides space for mechanical lines within a building (e.g., HVAC, plumbing, etc.). The active space **2** provides room for the building occupants during normal intended use of the building (e.g., in an office building, the active space would be occupied by offices containing computers, lamps, etc.). The plenary space **3** may exist above the first building panels **100** and the active room environment **2** may exist below the second building panels **200**.

(28) Referring to FIGS. 3B and 4B, other embodiments of the present invention include a building system **1** comprising a plurality of the second building panels **200** and one or more support

elements **50** (also referred to as a “support member”)—as discussed in greater detail herein.

(29) Referring now to FIG. **1**, the first building panel **100** (also referred to as a “backing building panel”) may comprise a first major surface **101** opposite a second major surface **102** and a side surface **103** extending between the first and second major surfaces **101**, **102**.

(30) The side surface **103** of the first building panel **100** may define a perimeter of the first building panel **100**. The side surface **103** of the first building panel **100** may define a perimeter of the first major surface **101** of the first building panel **100**. The side surface **103** of the first building panel **100** may define a perimeter of the second major surface **102** of the first building panel **100**.

(31) Depending on the shape of the first building panel **100**, there may be a plurality of side surfaces **103**. The first building panel **100** may be rectilinear in shape. The first building panel **100** may be elongated and extend along a first longitudinal axis A-A.

(32) According to such embodiments, the side surface **103** of the first building panel **100** may comprise a first side surface **103a**, a second side surface **103b**, a third side surface **103c**, and a fourth side surface **103d**. The first side surface **103a** may be substantially parallel to the second side surface **103b**. The third side surface **103c** may be substantially parallel to the fourth side surface **103d**. The first side surface **103a** and the second side surface **103b** may intersect the third side surface **103c** and the fourth side surface **103d**. The first side surface **103a** and the second side surface **103b** may each be substantially orthogonal to each of the third side surface **103c** and the fourth side surface **103d**.

(33) The first longitudinal axis A-A may intersect the third side surface **103c** and the fourth side surface **103d**. The first longitudinal axis A-A may extend substantially parallel to the first side surface **103a** and the second side surface **103b**. The first longitudinal axis A-A may extend substantially orthogonal to the third side surface **103c** and the fourth side surface **103d**.

(34) The first side surface **103a** may be referred to as a first elongated side surface or a first elongated edge. The second side surface **103b** may be referred to as a second elongated side surface or a second elongated edge. The third side surface **103c** may be referred to as a first transverse side surface or a first transverse edge. The fourth side surface **103d** may be referred to as a second transverse side surface or a second transverse edge.

(35) The first building panel **100** may have a first thickness $t_{sub.1}$ as measured from the first major surface **101** to the second major surface **102** of the first building panel **100**. The first thickness $t_{sub.1}$ may range from about 0.3 inches to about 0.75 inches—including all thicknesses and sub-ranges there-between. In some embodiments, the first thickness $t_{sub.1}$ may range from about 0.375 inches to about 0.625 inch—including all lengths and sub-ranges there-between.

(36) The first building panel **100** may have a first length $L_{sub.1}$ as measured from the third side surface **103c** to the fourth side surface **103d**. The first length $L_{sub.1}$ may range from about 84 inches to about 156 inches—including all lengths and sub-ranges there-between. In some embodiments, the first length $L_{sub.1}$ may range from about 96 inches to about 144 inches—including all lengths and sub-ranges there-between. In some embodiments, the first length $L_{sub.1}$ is about 96 inches. In some embodiments, the first length $L_{sub.1}$ is about 120 inches.

(37) The first building panel **100** may have a first width $W_{sub.1}$ as measured from the first side surface **103a** to the second side surface **103b**. The first width $W_{sub.1}$ may range from about 36 inches to about 60 inches—including all widths and sub-ranges there-between. In some embodiments, the first width $W_{sub.1}$ may range from about 46 inches to about 50 inches—including all widths and sub-ranges there-between. In some embodiments, the first width $W_{sub.1}$ is about 48 inches.

(38) The first building panel **100** may comprise a first body **110**. The first body **110** may form the first major surface **101**, the second major surface **102**, and the side surface **103** of the first building panel **100**.

(39) The first body **110** may be a non-porous body. The term non-porous refers to the first body **110** preventing airflow through the first building panel **100** between the first major surface **101** and the

second major surface **102** under atmospheric conditions.

(40) The first building panel **100** may exhibit a CAC value of 30 or greater, preferably 35 or greater.

(41) The first body **110** of the first building panel **100** may be formed of gypsum, plywood, particle board, drywall, or the combinations thereof.

(42) The first building panel **100** may have a first density. The first density may range from about 16 lb./ft.^{sup.3} to about 180 lb./ft.^{sup.3}—including all densities and sub-ranges there-between. In some embodiments, the first density may range from about 25 lb./ft.^{sup.3} to about 100 lb./ft.^{sup.3}—including all densities and sub-ranges there-between.

(43) Referring now to FIGS. 2A and 2B, the second building panel **200** (also referred to as a “facing building panel”) may comprise a first major surface **201** opposite a second major surface **202** and a side surface **203** extending between the first and second major surfaces **201**, **202**.

(44) The side surface **203** of the second building panel **200** may define a perimeter of the second building panel **200**. The side surface **203** of the second building panel **200** may define a perimeter of the first major surface **201** of the second building panel **200**. The side surface **203** of the second building panel **200** may define a perimeter of the second major surface **202** of the second building panel **200**.

(45) Depending on the shape of the second building panel **200**, there may be a plurality of side surfaces **203**. The second building panel **200** may be rectilinear in shape. The second building panel **200** may be elongated and extend along a second longitudinal axis B-B.

(46) According to such embodiments, the side surface **203** of the second building panel **200** may comprise a first side surface **203a**, a second side surface **203b**, a third side surface **203c**, and a fourth side surface **203d**. The first side surface **203a** may be substantially parallel to the second side surface **203b**. The third side surface **203c** may be substantially parallel to the fourth side surface **203d**. The first side surface **203a** and the second side surface **203b** may intersect the third side surface **203c** and the fourth side surface **203d**. The first side surface **203a** and the second side surface **203b** may each be substantially orthogonal to each of the third side surface **203c** and the fourth side surface **203d**.

(47) The second longitudinal axis B-B may intersect the third side surface **203c** and the fourth side surface **203d**. The second longitudinal axis B-B may extend substantially parallel to the first side surface **203a** and the second side surface **203b**. The second longitudinal axis B-B may extend substantially orthogonal to the third side surface **203c** and the fourth side surface **203d**.

(48) The first side surface **203a** may be referred to as a first elongated side surface or a first elongated edge. The second side surface **203b** may be referred to as a second elongated side surface or a second elongated edge. The third side surface **203c** may be referred to as a first transverse side surface or a first transverse edge. The fourth side surface **203d** may be referred to as a second transverse side surface or a second transverse edge.

(49) The second building panel **200** may have a second thickness $t_{\text{sub.2}}$ as measured from the first major surface **201** to the second major surface **202** of the second building panel **200**. The second thickness $t_{\text{sub.2}}$ may range from about 0.4 inches to about 1.6 inches—including all thicknesses and sub-ranges there-between. In some embodiments, the second thickness $t_{\text{sub.2}}$ may range from about 0.75 inches to about 1.0 inch—including all lengths and sub-ranges there-between. In some embodiments, the second thickness $t_{\text{sub.2}}$ is about 0.88 inches.

(50) The second building panel **200** may have a second length $L_{\text{sub.2}}$ as measured from the third side surface **203c** to the fourth side surface **203d**. The second length $L_{\text{sub.2}}$ may range from about 60 inches to about 84 inches—including all lengths and sub-ranges there-between. In some embodiments, the second length $L_{\text{sub.2}}$ may range from about 70 inches to about 74 inches—including all lengths and sub-ranges there-between. In some embodiments, the second length $L_{\text{sub.2}}$ is about 72 inches.

(51) The second building panel **200** may have a second width $W_{\text{sub.2}}$ as measured from the first

side surface **203a** to the second side surface **203b**. The second width W.sub.2 may range from about 36 inches to about 60 inches—including all widths and sub-ranges there-between. In some embodiments, the second width W.sub.2 may range from about 46 inches to about 50 inches—including all widths and sub-ranges there-between. In some embodiments, the second width W.sub.2 is about 48 inches.

(52) The first major surface **201** of the second building panel **200** may comprise a central region R.sub.C and a perimeter region R.sub.P. The central region R.sub.C may be at least partially circumscribed by the perimeter region R.sub.P. The central region R.sub.C may be fully circumscribed by the perimeter region R.sub.P. The perimeter region R.sub.P may extend from the central region R.sub.C to the perimeter of the second building panel **200** (the perimeter of the second building panel **200** being located at the intersection of the side surface **203** and the first major surface **201** of the second building panel **200**). The perimeter region R.sub.P may extend from the central region R.sub.C to the perimeter of the second building panel **200** by a perimeter distance D.sub.5.

(53) The perimeter distance D.sub.5 may range from about 0.5 inches to about 6.0 inches—including all distances and sub-ranges there-between. According to the embodiments, of FIGS. **3A** and **4A**, the perimeter distance D.sub.5 may range from about 3.5 inches to about 4.5 inches—including all distances and sub-ranges there-between. According to the embodiments, of FIGS. **3B** and **4B**, the perimeter distance D.sub.5 may range from about 0.5 inches to about 1.0 inches—including all distances and sub-ranges there-between.

(54) The first major surface **201** of the second building panel **200** may occupy a first surface area. The central region R.sub.C of the first major surface **201** of the second building panel **200** may occupy a second surface area. The perimeter region R.sub.P of the first major surface **201** of the second building panel **200** may occupy a third surface area. The summation of the second surface area and the third surface area may be substantially equal to the first surface area. Stated otherwise, the third surface area may be equal to the difference between the first surface area and the second surface area.

(55) The second surface area may range from about 3% to about 30% of the first surface area—including all percentages and sub-ranges there-between. According to the embodiments, of FIGS. **3A** and **4A**, the second surface area may range from about 20% to about 30% of the first surface area—including all percentages and sub-ranges there-between. According to the embodiments, of FIGS. **3A** and **4A**, the second surface area may range from about 24% to about 26% of the first surface area—including all percentages and sub-ranges there-between. According to the embodiments, of FIGS. **3B** and **4B**, the second surface area may range from about 3% to about 10% of the first surface area—including all percentages and sub-ranges there-between. According to the embodiments, of FIGS. **3B** and **4B**, the second surface area may range from about 3% to about 7% of the first surface area—including all percentages and sub-ranges there-between.

(56) The second building panel **200** may comprise a second body **210**. In some embodiments, the second building panel **200** may comprise a non-woven scrim **220**. The non-woven scrim **220** may be bonded to the second body **210** such that the non-woven scrim **220** forms the first major surface **201** of the second building panel **200**. The second body **210** may form the second major surface **202** of the second building panel **200**. The second body **210** may form the side surface **203** of the second building panel **200**.

(57) The second body **210** may be a fibrous body. The second body **210** may be comprised of a binder and fibers. In some embodiments, the second body **210** may further comprise a filler and/or additive. Non-limiting examples of filler may include powders of calcium carbonate, limestone, titanium dioxide, sand, barium sulfate, clay, mica, dolomite, silica, talc, perlite, polymers, gypsum, wollastonite, expanded-perlite, calcite, aluminum trihydrate, pigments, zinc oxide, or zinc sulfate.

(58) The fibers may be organic fibers, inorganic fibers, or a blend thereof. Non-limiting examples of inorganic fibers mineral wool (also referred to as slag wool), rock wool, stone wool, and glass

fibers. Non-limiting examples of organic fiber include fiberglass, cellulosic fibers (e.g. paper fiber—such as newspaper, hemp fiber, jute fiber, flax fiber, wood fiber, or other natural fibers), polymer fibers (including polyester, polyethylene, aramid—i.e., aromatic polyamide, and/or polypropylene), protein fibers (e.g., sheep wool), and combinations thereof.

(59) Non-limiting examples of binder may include a starch-based polymer, polyvinyl alcohol (PVOH), a latex, polysaccharide polymers, cellulosic polymers, protein solution polymers, an acrylic polymer, polymaleic anhydride, epoxy resins, or a combination of two or more thereof.

(60) The second body **210** may be porous, thereby allowing airflow through the second body **210** between the first major surface **201** and the second major surface **202** of the second body **210**.

According to the present invention, the term porous refers to the second body **210** having open pathways sufficient for enough airflow through the second body **210** (under atmospheric conditions) for the second body **210** and the resulting second building panel **200** to function as an acoustic building panel and corresponding acoustic building system **1**, which requires properties related to noise reduction and sound attenuation properties—as discussed further herein.

(61) Specifically, the second body **210** of the present invention may have a porosity ranging from about 60% to about 98%—including all values and sub-ranges there between. In a preferred embodiment, the second body **210** has a porosity ranging from about 75% to 95%—including all values and sub-ranges there between.

(62) According to the embodiments where the second body **210** is formed from binder and fibers, porosity may be calculated by the following:

$$\% \text{ Porosity} = [V.\text{sub.Total} - (V.\text{sub.Binder} + V.\text{sub.F} + V.\text{sub.A} + V.\text{sub.Filler})] / V.\text{sub.Total}$$

(63) Where V.sub.Total refers to the total volume of second body **210** defined by a first major surface of the second body **210**, a second major surface of second body **210**, and a side surface of second body **210**. V.sub.Binder refers to the total volume occupied by the binder in the second body **210**. V.sub.F refers to the total volume occupied by the fibers in the second body **210**. V.sub.Filler refers to the total volume occupied by the filler in the second body **210**. V.sub.A refers to the total volume occupied by additives in the second body **210**. Thus, the % porosity represents the amount of free volume within the second body **210**.

(64) The second body **210** of the present invention may exhibit sufficient airflow for the second building panel **200** to have the ability to reduce the amount of reflected sound in an active room environment **2**. The reduction in amount of reflected sound in an active room environment **2** is expressed by a Noise Reduction Coefficient (NRC) rating as described in American Society for Testing and Materials (ASTM) test method C423. This rating is the average of sound absorption coefficients at four ¼ octave bands (250, 500, 1000, and 2000 Hz), where, for example, a system having an NRC of 0.90 has about 90% of the absorbing ability of an ideal absorber. A higher NRC value indicates that the material provides better sound absorption and reduced sound reflection.

(65) The second body **210** of the present invention exhibits an NRC of at least about 0.5. In a preferred embodiment, the second body **210** of the present invention may have an NRC ranging from about 0.60 to about 0.99—including all value and sub-ranges there-between. The second building panel **200** of the present invention exhibits an NRC of at least about 0.5. In a preferred embodiment, the second building panel **200** of the present invention may have an NRC ranging from about 0.60 to about 0.99—including all value and sub-ranges there-between.

(66) In addition to reducing the amount of reflected sound in a single active room environment **2**, the second body **210**—and resulting second building panel **200**—of the present invention may also be able to exhibit superior sound attenuation—which is a measure of the sound reduction between an active room environment **2** and a plenary space **3**. The ASTM has developed test method E1414 to standardize the measurement of airborne sound attenuation between room environments **2** sharing a common plenary space **3**. The rating derived from this measurement standard is known as the Ceiling Attenuation Class (CAC). Ceiling materials and systems having higher CAC values have a greater ability to reduce sound transmission through the plenary space **3**—i.e. sound

attenuation function.

(67) The second body **210** of the present invention may exhibit a CAC value of 30 or greater, preferably 35 or greater. The second building panel **200** of the present invention may exhibit a CAC value of 30 or greater, preferably 35 or greater.

(68) The second building panel **200** of the present invention may have a second density. The second density may range from about 2 lb./ft.^{sup.3} to about 16 lb./ft.^{sup.3}—including all densities and sub-ranges there-between. In some embodiments, the second density may range from about 5 lb./ft.^{sup.3} to about 14 lb./ft.^{sup.3}—including all densities and sub-ranges there-between.

(69) The first density and the second density may not be equal. A ratio of the first density to the second density may range from about 1.5:1 to about 10:1—including all densities and sub-ranges there-between. In some embodiments, the ratio of the first density to the second density may be at least 2:1, preferably 3:1. In some embodiments, the ratio of the first density to the second density may be about 4:1.

(70) The first thickness t.sub.1 of the first building panel **100** and the second thickness t.sub.2 of the second building panel **200** may not be equal. A ratio of the second thickness t.sub.2 to the first thickness t.sub.1 may range from about 3.0:1 to about 1.25:1—including all ratios and sub-ranges there-between. In some embodiments, the ratio of the second thickness t.sub.2 to the first thickness t.sub.1 may be about 2.3:1. In some embodiments, the ratio of the second thickness t.sub.2 to the first thickness t.sub.1 may be about 1.8:1. In some embodiments, the ratio of the second thickness t.sub.2 to the first thickness t.sub.1 may be about 1.4:1.

(71) A ratio of the first length L.sub.1 to the second length L.sub.2 may range from about 3.0:1 to about 1.2:1—including all ratios and sub-ranges there-between. In some embodiments, the ratio of the first length L.sub.1 to the second length L.sub.2 may be about 2:1. In some embodiments, the ratio of the first length L.sub.1 to the second length L.sub.2 may be about 1.7:1. In some embodiments, the ratio of the first length L.sub.1 to the second length L.sub.2 may be about 1.3:1.

(72) Referring now to FIGS. **3A**, **4A**, **5A**, **6**, **7**, **8A**, and **8B** a building system **1** may comprise a plurality of the first building panels **100** and a plurality of the second building panels **200**. Each of the plurality of first building panels **100** may be secured to the support member **50**. Each of the plurality of first building panels **100** may be secured directly to the support member **50**.

Specifically, each of the plurality of first building panels **100** may be secured to the support face **51** of the support member **50**. Each of the plurality of first building panels **100** may be directly secured to the support face **51** of the support member **50**.

(73) Each of the plurality of first building panels **100** may be secured to the support member **50** by a mechanical fastener **400**. In other embodiments, each of the plurality of first building panels **100** may be secured to the support member **50** by an adhesive (not pictured).

(74) During installation, each of the plurality of first building panels **100** may be positioned such that the second major surface **102** of the first building panel **100** faces the support surface **51** of the support member **50**. In the embodiment shown in FIGS. **3** and **4**, the second major surface **102** of the first building panel **100** faces upward toward the plenary space **3**. According to embodiments where the building system is a wall system, the second major surface **102** may be horizontally oriented and face away from the active room environment **2** (not pictured).

(75) During installation, each of the plurality of first building panels **100** may be positioned such that the first major surface **101** of the first building panel **100** faces away from the support surface **51** of the support member **50**. In the embodiment shown in FIGS. **3** and **4**, the first major surface **101** of the first building panel **100** faces downward toward the active room environment **2**.

According to embodiments where the building system is a wall system, the first major surface **101** may be horizontally oriented and face toward from the active room environment **2** (not pictured).

(76) A first interface **60** may exist between the overlapping space that exists between the second major surface **102** of the first building panel **100** and the support surface **51** of the support member **50**. The first interface **60** may comprise adhesive **300**. One or more of the mechanical fastener **400**

may extend through the first interface **60**.

(77) In some embodiments, the first interface **60** may be substantially free of adhesive. The term “substantially free of adhesive” refers to an amount of adhesive present in the first interface **60** that is not sufficient for the adhesive to physically bond the first building panel **100** to the support member **50** under the effects of gravity. Therefore, the term “substantially free of adhesive” may include a first interface **60** that is entirely free of adhesive or may include minor amounts of adhesive that is not sufficient alone to physically secure the first building panel **100** to the support member **50**. In some embodiments, the first interface **60** may be entirely free of adhesive.

(78) Once secured to the support members **50**, the plurality of first building panels **100** may form a pre-existing building layer **80**. Each of the plurality of second building panels **200** may be secured to the pre-existing building layer **80** by one or more fasteners **400**. Each of the plurality of second building panels **200** may be secured directly to the pre-existing building layer **80**. Specifically, each of the plurality of second building panels **200** may be secured to one or more of the first building panel **100**. Each of the plurality of second building panels **200** may be directly secured to one or more of the first building panels **100**.

(79) While the present invention provides for the installation and formation of the pre-existing building layer **80**, the present invention is not limited to such installation steps. Rather, other embodiments of the present invention may not require the separate step of installing the plurality of first building panels **100** as the pre-existing building layer **80** may already be present in a room environment—according to such embodiments, the plurality of second building panels **200** may be retro-fit to a pre-existing building layer **80**.

(80) Each of the plurality of second building panels **200** may be secured to the pre-existing building layer **80** by mechanical fastener **400**. Each of the plurality of second building panels **200** may be secured to the pre-existing building layer **80** by adhesive **300**. Each of the plurality of second building panels **200** may be secured to one or more of the first building panels **100** by mechanical fastener **400**. Each of the plurality of second building panels **200** may be secured to one or more of the first building panels **100** by adhesive **300**. The plurality of second building panels **200** installed adjacent to each other may form a facing layer **90**.

(81) In the installed state, each of the plurality of second building panels **200** may be positioned such that the second major surface **202** of the second building panel **200** faces the first major surface **101** of the first building panels **100**. In the embodiment shown in FIGS. 3A and 4A, the second major surface **202** of the second building panel **200** faces upward. According to embodiments where the building system is a wall system, the second major surface **202** of the second building panel **200** may be horizontally oriented and face away from the active room environment **2** (not pictured).

(82) In the installed state, each of the plurality of second building panels **200** may be positioned such that the first major surface **201** of the second building panel **200** faces away from the support surface **51** of the support member **50**. In the embodiment shown in FIGS. 3A and 4A, the first major surface **201** of the second building panel **300** faces downward toward the active room environment **2**. According to embodiments where the building system is a wall system, the first major surface **201** may be horizontally oriented and face toward from the active room environment **2** (not pictured).

(83) A second interface **70** may exist between the overlapping space that exists between the second major surface **202** of the second building panel **200** and the first major surface **101** of the first building panel **100**. The second interface **70** may comprise adhesive. One or more of the mechanical fastener **400** may extend through the second interface **70**.

(84) The mechanical fasteners **400** may comprise a plurality of first mechanical fasteners **410** and a plurality of second mechanical fastener **420**. The plurality of second building panels **200** may be secured to the pre-existing building layer **80** by the plurality of first mechanical fasteners **410**. The plurality of second building panels **200** may be secured to one or more of the first building panels

200 by the plurality of first mechanical fasteners **410**. The plurality of first building panels **100** may be secured to the securing members **50** by the plurality of second mechanical fasteners **420**.

(85) Each of the plurality of first mechanical fasteners **410** may comprise a head **411** and a threaded body **412**. Each of the plurality of first mechanical fasteners **410** may extend a first fastener length L.sub.F1 as measured from a proximal end **413** of the head **411** to a distal end **414** of the threaded body **412** of the first mechanical fastener **410**. The head **411** of each of the first mechanical fasteners **410** may have an outer diameter D.sub.4.

(86) Each of the plurality of second mechanical fasteners **420** may comprise a head **421** and a threaded body **422**. Each of the plurality of second mechanical fasteners **420** may extend a second fastener length that is measured from a proximal end **423** of the head **421** to a distal end **424** of the threaded body **422** of the second mechanical fastener **420**. The second fastener length may be substantially equal to the first fastener length L.sub.F1. In some embodiments, the second fastener length may be greater than the first fastener length L.sub.F1.

(87) The first fastener length L.sub.F1 may range from about 1 inch to about 2.5 inches—including all lengths and sub-ranges there-between. In some embodiments, the first fastener length L.sub.F1 may range from about 1 inch to about 2 inches—including all lengths and sub-ranges there-between.

(88) The first fastener length L.sub.F1 may be less than the summation of the first thickness t.sub.1 of the first building panel and the second thickness t.sub.2 of the second building panel t.sub.2. In some embodiments, the first fastener length L.sub.F1 may be substantially equal to the summation of the first thickness t.sub.1 of the first building panel and the second thickness t.sub.2 of the second building panel t.sub.2. In some embodiments, the first fastener length L.sub.F1 may be greater than the summation of the first thickness t.sub.1 of the first building panel and the second thickness t.sub.2 of the second building panel t.sub.2.

(89) A ratio of the first fastener length L.sub.F1 to the first thickness t.sub.1 of the first building panel **100** may range from about 1.25:1 to about 2.5:1—including all ratios and sub-ranges there-between. In some embodiments, the ratio of the first fastener length L.sub.F1 to the first thickness t.sub.1 of the first building panel **100** may range from about 1.5:1 to about 2.0:1—including all ratios and sub-ranges there-between.

(90) The outer diameter D.sub.4 of the head **411** of the first mechanical fastener **410** may range from about 0.3 inch to about 0.38 inches—including all diameters and sub-ranges there-between. In some embodiments, the outer diameter D.sub.4 of the head **411** of the first mechanical fastener **410** may range from about 0.31 inch to about 0.35 inches—including all diameters and sub-ranges there-between.

(91) A ratio of the first fastener length L.sub.F1 to the outer diameter D.sub.4 of the first mechanical fastener **410** may range from about 6:1 to about 4:1—including all ratios and sub-ranges there-between. In some embodiments, the ratio of the first fastener length L.sub.F1 to the outer diameter D.sub.4 of the first mechanical fastener **410** may range from about 5.5:1 to about 4.25:1—including all ratios and sub-ranges there-between.

(92) In securing the second building panel **200** to the pre-existing building layer **80**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that the head **411** of the first fastener **410** is located inset of the first major surface **201** of the second building panel **200**. State otherwise, in securing the second building panel **200** to one or more of the first building panel **100**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that the head **411** of the first fastener **410** is located inset of the first major surface **201** of the second building panel **200**.

(93) At least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that the head **411** is entirely positioned between the first major surface **201** and the second major surface **202** of the second building panel **200**. At least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that the head **411** of the first fastener

410 does not overlap with the second major surface **201** of the second building panel **200**. At least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that the proximal end of the head **411** is inset from the first major surface **201** of the second building panel **200** by a first depth D.sub.1. The first depth D.sub.1 may range from about 1% to about 30% the second thickness t.sub.2 of the second building panel **200**—including all percentages and sub-ranges there-between.

(94) In securing the second building panel **200** to the pre-existing building layer **80**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that at least a portion of the threaded body **412** of the first fastener **410** is located inset of the first major surface **201** of the second building panel **200**. In securing the second building panel **200** to the pre-existing building layer **80**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that at least a portion of the threaded body **412** of the first fastener **410** is located between the first major surface **201** and the second major surface **202** of the second building panel **200**.

(95) In securing the second building panel **200** to one or more of the first building panel **100**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that at least a portion of the threaded body **412** of the first fastener **410** is located inset of the first major surface **201** of the second building panel **200**. In securing the second building panel **200** to one or more of the first building panel **100**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that at least a portion of the threaded body **412** of the first fastener **410** is located between the first major surface **201** and the second major surface **202** of the second building panel **200**.

(96) In securing the second building panel **200** to one or more of the first building panel **100**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that at least a portion of the threaded body **412** passes through adhesive **300** present in the second interface **70**. According to such embodiments, the first fastener **410** may bring together the first building panel **100** and the second building panel **200** where the adhesive **300** is located, thereby spreading the adhesive **300** within the second interface **70** to ensure a strong adhesive bond therein.

(97) In securing the first building panel **100** to the securing member **50**, at least one of the plurality of second fasteners **420** may be driven into the first building panel **100** such that the head **421** of the second fastener **420** is at least substantially flush with the first major surface **101** of the first building panel **100**. In some embodiments, in securing the first building panel **100** to the securing member **50**, at least one of the plurality of second fasteners **420** may be driven into the first building panel **100** such that the head **421** of the second fastener **420** is inset of the first major surface **101** of the first building panel **100**.

(98) In securing the first building panel **100** to the securing member **50**, at least one of the plurality of second fasteners **420** may be driven into the first building panel **100** such that at least a portion of the threaded body **422** of the second fastener **420** is located inset of the first major surface **101** of the first building panel **100**. In securing the first building panel **100** to the securing member **50**, at least one of the plurality of second fasteners **420** may be driven into the first building panel **100** such that at least a portion of the threaded body **422** of the second fastener **420** is located between the first major surface **101** and the second major surface **102** of the first building panel **100**.

(99) In securing the first building panel **100** to one or more of the securing members **50**, at least one of the plurality of second fasteners **420** may be driven into the securing member **50** such that at least a portion of the threaded body **422** of the second fastener **420** is located inset of the support surface **51** of the securing member **50**.

(100) At least a portion of the threaded body **422** of one of the plurality of the second fasteners **420** may extend through the first interface **60**. In some embodiments, the threaded body **422** of at least one of the plurality of second fasteners **420** may not extend through the second interface **70**.

(101) The plurality of second fasteners **420** may be located inset of the side surfaces **103** of the first building panel **100**—whereby the plurality of second mechanical fasteners **420** contact a single one of the first building panel **100**.

(102) At least a portion of the threaded body **412** of one of the plurality of the first fasteners **410** may extend through the second interface **70**. In some embodiments, the threaded body **412** of at least one of the plurality of first fasteners **410** may not extend through the first interface **60**.

(103) During installation, the second major surface **202** of at least one of the second building panel **200** is brought into contact with a pre-existing building surface **81** formed by the pre-existing building layer **80** to form the second interface **70**, whereby the adhesive **300** is present as the second interface **70** is formed, and the adhesive is in a state where the adhesive **300** has not yet reach full bond strength.

(104) The adhesive may be a reactive adhesive provided in an uncured state—i.e., has not yet reached full bond strength. Non-limiting examples of reactive adhesives include epoxy, polyurethane, and cyanoacrylates—whereby the uncured adhesive may be applied and allowed to react to reach a fully cured state, thereby reaching full bond strength. The adhesive may be a non-reactive adhesive, which may be applied in a first state that has not yet reached full bond strength and allowed to transition into a second state that exhibits full bond strength. Non-limiting examples of non-reactive adhesives include drying adhesives (solvent-based), hot-melt adhesives, and contact adhesives. Drying adhesives may be applied in a wet-state (i.e., first state) and allowed to dry (i.e., second state) to reach full bond strength. Hot-melt adhesives may be applied in a melted-state (i.e., first state) and allowed to cool (i.e., second state) to reach full bond strength.

(105) Installation may further comprise driving at least one of the first mechanical fasteners **410** into the at least one second building panel **200** and the first building panel **100**. The first mechanical fastener **410** may be driven into the first and second building panels **100**, **200** and through the second interface **70** before the adhesive **300** has reached full bond strength.

Subsequently, installation may include a time period sufficient for such adhesive **300** to reach full bond strength. After the time period where the adhesive **300** reaches full bond strength, one or more of the plurality of first mechanical fasteners **410** may be removed.

(106) Referring now to FIG. 7, the thin dotted linear lines represent the support member **50** and the solid lines represent the first building panels **100**—the support members **50** are represented by dotted lines so that it may be shown how the plurality of first building panels **100** are oriented relative to the plurality of support members **50** in forming the pre-existing building layer **80**.

(107) Referring now to FIGS. 8A and 8B, the thin dotted linear lines of the same length represent the plurality of first building panels **100**, the thin solid linear lines represent the second building panels **200**, the thin solid wavy lines represent the adhesive **300**, and the dotted-dashed linear lines represent the boundary between the central region R.sub.C and the perimeter region R.sub.P of each second building panel **200**. The first building panels **100** are represented by as thin dotted lines so that it may be shown how the plurality of second building panels **200** are oriented relative to the plurality of first building panels **100** in forming the facing layer **90** as well as the orientation of the adhesive **300** within the second interface **70** located between the pre-existing building layer **80** and the facing layer **90**.

(108) When installed, the plurality of second building panels **200** and the plurality of first building panels **100** may be arranged in an offset pattern such that at least one of the side surfaces **103** of the first building panel and at least one of the side surfaces **203** of the second building panel are not coplanar.

(109) For each one of the plurality of second building panels **200**, the plurality of first mechanical fasteners **410** may be separated by a longitudinal separation distance D.sub.2 and a lateral separation distance D.sub.3. The longitudinal separation distance D.sub.2 is the distance separating the adjacent most first mechanical fasteners along a direction parallel to the second longitudinal axis B-B. The lateral separation distance D.sub.3 is the distance separating the adjacent most first

mechanical fasteners along a direction orthogonal to the second longitudinal axis B-B.

(110) When installed, at least one of the plurality of first building panels **100** and at least one of the plurality of second building panels **200** are arranged such that the first longitudinal axis A-A and the second longitudinal axis B-B are not parallel. When installed, at least one of the plurality of first building panels **100** and at least one of the plurality of second building panels **200** are arranged such that the first longitudinal axis A-A and the second longitudinal axis B-B are orthogonal.

(111) When installed, the first elongated edge **203a** of one of the plurality of second building panels **200** may overlap at least a portion of a first one of the plurality of first building panels **100** and a portion of a second one of the plurality of first building panels **100**. When installed, the second elongated edge **203b** of the one of the plurality of second building panels **200** may overlap at least the portion of the first one of the plurality of first building panels **100** and a portion of a third one of the plurality of first building panels **100**.

(112) During installation of the second building panels **200**, at least one of the first and/or third mechanical fasteners **410**, **430** are driven into the edge interface **40** and/or the second building panel **200** before the adhesive **300** has reached full bond strength. Subsequently, installation may include a time period sufficient for such adhesive **300** to reach full bond strength. After the time period where the adhesive **300** reaches full bond strength, one or more of the plurality of third mechanical fasteners **430** may be removed.

(113) For the adhesive **300** present in the second interface **70**, the adhesive **300** may be present in longitudinal stripes that are substantially parallel to the second longitudinal axis B-B. For each second building panel **200**, there may be at least two stripes of adhesive **300** extending in a direction substantially parallel to the longitudinal axis B-B. Each of the strips of adhesive **300** may be separated by a distance that is substantially equal to the lateral separation distance D.sub.3 of the first mechanical fasteners **410**.

(114) The plurality of longitudinal strips of the adhesive **300** may continuously span across at least two adjacent ones of the second building panel **200**. The plurality of longitudinal stripes of the adhesive **300** may continuously span across at least two adjacent ones of the second building panel **200** in a direction substantially parallel to the second longitudinal axis B-B. The plurality of longitudinal stripes of the adhesive **300** may continuously span across at least three adjacent ones of the second building panel **200** in a direction substantially parallel to the second longitudinal axis B-B.

(115) The plurality of longitudinal stripes of the adhesive **300** may continuously span across at least two adjacent ones of the first building panel **100**. The plurality of longitudinal stripes of the adhesive **300** may continuously span across at least two adjacent ones of the first building panel **100** in a direction substantially orthogonal to the first longitudinal axis A-A.

(116) For the adhesive **300** present in the second interface **70**, the adhesive **300** may be present in longitudinal strips that are substantially parallel to the second longitudinal axis B-B. For each second building panel **200**, there may be at least two stripes of adhesive **300** extending in a direction substantially parallel to the longitudinal axis B-B. Each of the strips of adhesive **300** may be separated by a distance that is substantially equal to the lateral separation distance D.sub.3 of the first mechanical fasteners **410**.

(117) The plurality of stripes of the adhesive **300** may continuously span across at least two adjacent ones of the second building panel **200**. The plurality of stripes of the adhesive **300** may continuously span across at least two adjacent ones of the second building panel **200** in a direction substantially parallel to the second longitudinal axis B-B. The plurality of stripes of the adhesive **300** may continuously span across at least three adjacent ones of the second building panel **200** in a direction substantially parallel to the second longitudinal axis B-B.

(118) The plurality of stripes of the adhesive **300** may continuously span across at least two adjacent ones of the first building panel **100**. The plurality of stripes of the adhesive **300** may continuously span across at least two adjacent ones of the first building panel **100** in a direction

substantially orthogonal to the first longitudinal axis A-A.

(119) Referring to FIG. 8A, the second building panel **200** may be secured such that a first plurality of the first mechanical fasteners **410a** are located within the central region R.sub.C of the second building panel **200** and a second plurality of the first mechanical fasteners **410b** are located within the perimeter region R.sub.P of the second building panel **200**.

(120) According to such embodiments, the perimeter region R.sub.P may have a perimeter distance ranging from about 3 inches to about 5 inches—including all distances and subranges there-between. According to such embodiments, the first plurality of first mechanical fasteners **410a** may be present in the central region R.sub.C an amount ranging from about 4 to about 10—including all integers and sub-ranges there-between. In some embodiments, the first plurality of first mechanical fasteners **410a** may be present in the central region R.sub.C an amount ranging from about 4 to about 8—including all integers and sub-ranges there-between. According to such embodiments, the first plurality of first mechanical fasteners **410a** may have a longitudinal separation distance D.sub.2 ranging from about 15 inches to about 21 inches—including all distances and subranges there-between. In some embodiments, the first plurality of first mechanical fasteners **410a** may have a longitudinal separation distance D.sub.2 of about 16 inches to about 18 inches.

(121) The first plurality of first mechanical fasteners **410a** may have a longitudinal separation distance D.sub.2 that is equal to about 20% to about 30% the second length L.sub.2 of the second building panel **200**—including all percentages and sub-ranges there-between.

(122) According to such embodiments, the second plurality of first mechanical fasteners **410b** may be present in the perimeter region R.sub.P an amount such that the second plurality of first mechanical fasteners **410b** are separated by longitudinal separation distance D.sub.2 and a lateral separation distance D.sub.3 ranging from about 6 inches to about 15 inches—including all distances and sub-ranges there-between. The second plurality of first mechanical fasteners **410b** may have a longitudinal separation distance D.sub.2 that is equal to about 8% to about 15% the second length L.sub.2 of the second building panel **200**—including all percentages and sub-ranges there-between.

(123) According to such embodiments, where adhesive **300** is present in the second interface **70**, the first plurality of first mechanical fasteners **410a** and the second plurality of first mechanical fasteners **410b** should extend through the adhesive **300**.

(124) The first plurality of first mechanical fastener **410a** may be present in a first fastener density. The first fastener density may be calculated as the total number of the first plurality of first mechanical fasteners **410a** that are located within the first surface area of the first major surface **201** of the second building panel **200**. The second plurality of first mechanical fastener **410b** may be present in a second fastener density. The second fastener density may be calculated as the total number of the second plurality of first mechanical fasteners **410b** that are located within the first surface area of the first major surface **201** of the second building panel **200**.

(125) The first fastener density and the second fastener density may not be equal. The second fastener density may be greater than the first fastener density. The first fastener density may range from about 0.15 fasteners/ft.sup.2 to about 0.5 fasteners/ft.sup.2—including all densities and sub-ranges there-between. The second fastener density may range from about 0.6 fasteners/ft.sup.2 to about 1.3 fasteners/ft.sup.2—including all densities and sub-ranges there-between. A ratio of the second fastener density to the first fastener density may range from about 1.2:1 to about 8.6:1—including all ratios and sub-ranges there-between.

(126) Referring now to FIG. 8B, an alternative to the embodiment of FIG. 8A is shown, whereby the plurality of second building panels **200** and may be secured by a plurality of the first mechanical fasteners **410** and a plurality of a third mechanical fasteners **430**. According to this embodiment, the second building panel **200** may be secured such that the plurality of the first mechanical fasteners **410** are located within the central region R.sub.C of the second building panel **200** and a plurality of the third mechanical fasteners **430** along the edge interface **40** of adjacent

second building panels **200**.

(127) According to such embodiments, the perimeter region R.sub.P may be free of the first mechanical fastener **410** whereby only the central region R.sub.C comprises the first mechanical fastener **410**. In other embodiments, the perimeter region R.sub.P may comprise one or more of the first mechanical fastener **410**.

(128) For the third mechanical fasteners **430** located along the edge interface **40**, the lateral separation distance D.sub.3 may range from about 8 inches to about 30 inches—including all distances and sub-ranges there-between.

(129) Referring to FIG. **8B**, the building system **1** may comprise the second building panel **200** secured to the preexisting building layer **80** by both the plurality of first mechanical fasteners **410** and a plurality of third mechanical fasteners **430** as well as adhesive **300**.

(130) The third mechanical fastener **430** may comprise a head and a threaded body. Each of the plurality of third mechanical fasteners **430** may extend a third fastener length as measured from a proximal end of the head to a distal end of the threaded body of the third mechanical fastener **430**. The head of each of the third mechanical fasteners **430** may have an outer diameter.

(131) The third fastener length may range from about 1 inch to about 2.5 inches—including all lengths and sub-ranges there-between. The third fastener length may be less than the summation of the first thickness t.sub.1 of the first building panel and the second thickness t.sub.2 of the second building panel t.sub.2. In some embodiments, the third fastener length may be substantially equal to the summation of the first thickness t.sub.1 of the first building panel and the second thickness t.sub.2 of the second building panel t.sub.2. In some embodiments, the third fastener length may be greater than the summation of the first thickness t.sub.1 of the first building panel and the second thickness t.sub.2 of the second building panel t.sub.2.

(132) A ratio of the third fastener length to the first thickness t.sub.1 of the first building panel **100** may range from about 1.25:1 to about 2.5:1—including all ratios and sub-ranges there-between. In some embodiments, the ratio of the third fastener length to the first thickness t.sub.1 of the first building panel **100** may range from about 1.5:1 to about 2.0:1—including all ratios and sub-ranges there-between.

(133) The outer diameter of the head of the third mechanical fastener **430** may range from about 1.0 inch to about 1.5 inches—including all diameters and sub-ranges there-between. In some embodiments, the outer diameter of the head of the third mechanical fastener **430** may range from about 1.2 inch to about 1.3 inches—including all diameters and sub-ranges there-between.

(134) A ratio of the third fastener length to the outer diameter of the third mechanical fastener **430** may range from about 1:1.1 to about 1.5:1—including all ratios and sub-ranges there-between. In some embodiments, the ratio of the third fastener length to the outer diameter of the third mechanical fastener **430** may range from about 1.2:1 to about 1.4:1—including all ratios and sub-ranges there-between.

(135) At least a portion of the threaded body of one of the plurality of the third fasteners **430** may extend through the second interface **70**. In some embodiments, the threaded body of at least one of the plurality of third fasteners **430** may not extend through the first interface **60**.

(136) Referring now to FIGS. **9-11**, once the plurality of second building panels **200** are secured in place by mechanical fastener and adhesive **300**, a facing surface **91** of the facing layer **90** may be formed collectively by the first major surfaces of the plurality of second building panels **200**. The concealing composition **500** is applied where the plurality of first mechanical fasteners **410** and/or third mechanical fasteners **430**—thereby concealing such mechanical fasteners as well as concealing such edge interfaces **40**.

(137) According to the present invention, it has been discovered that, by securing the second building panels **200** to the pre-existing building layer **80** with at least the adhesive **300** and at least one of the first mechanical fastener **410** at the first depth D.sub.1, not only are fewer of the first mechanical fasteners **410** required to seismically secure such plurality of second building panels

200—thereby necessitating fewer first mechanical fasteners **410** to conceal on the facing surface **91**, but the concealment process is also simplified as the hole formed into the second building panel **200** by the first mechanical fastener **410** is more easily filled by the concealment compound **500**. Thus, the present invention provides a new and improvement seismically stable building surface that can be installed with better aesthetics and requiring less skilled labor during the concealment process.

(138) Subsequent to application of the concealment compound **500**, one or more coats of a paint composition **600** may be applied to the entirety of the facing surface **91** of the facing layer **90** comprising the concealment compound **500**. Once the facing layer **90** is formed after installing the plurality of second building panels **200**, a concealing composition **500** may be applied to a facing surface **91** of the facing layer **90**. Non-limiting examples of concealment compound **500** include spackle.

(139) Referring now to FIGS. **3B**, **4B**, **5B**, **6**, **7**, **8C**, **8D** a building system **1** may comprise a plurality of the second building panels **200** secured to the support member **50**. Each of the plurality of second building panels **200** may be secured directly to the support member **50**. Specifically, each of the plurality of second building panels **200** may be secured to the support face **51** of the support member **50**. Each of the plurality of second building panels **200** may be directly secured to the support face **51** of the support member **50**.

(140) Each of the plurality of second building panels **200** may be secured to the support member **50** by a mechanical fastener **400**. In other embodiments, each of the plurality of second building panels **200** may be secured to the support member **50** by the adhesive **300**.

(141) During installation, each of the plurality of second building panels **200** may be positioned such that the second major surface **202** of the second building panel **200** faces the support surface **51** of the support member **50**. In the embodiment shown in FIGS. **3B** and **4B**, the second major surface **202** of the second building panel **200** faces upward toward the plenary space **3**. According to embodiments where the building system is a wall system, the second major surface **202** may be horizontally oriented and face away from the active room environment **2** (not pictured).

(142) During installation, each of the plurality of second building panels **200** may be positioned such that the first major surface **201** of the second building panel **200** faces away from the support surface **51** of the support member **50**. In the embodiment shown in FIGS. **3B** and **4B**, the first major surface **201** of the second building panel **200** faces downward toward the active room environment **2**. According to embodiments where the building system is a wall system, the first major surface **201** may be horizontally oriented and face toward from the active room environment **2** (not pictured).

(143) A first interface **60** may exist between the overlapping space that exists between the second major surface **202** of the second building panel **200** and the support surface **51** of the support member **50**. The first interface **60** may comprise adhesive **300**. One or more of the mechanical fastener **400** may extend through the first interface **60**.

(144) Each of the plurality of second building panels **200** may be secured to the support member **50** by mechanical fastener **400**. Each of the plurality of second building panels **200** may be secured to the support member **50** by adhesive **300**. Each of the plurality of second building panels **200** may be secured to the support member **50** by the plurality of mechanical fastener **400** and the adhesive **300**. The plurality of second building panels **200** installed adjacent to each other may form a facing layer **90**.

(145) The mechanical fasteners **400** may comprise the plurality of first mechanical fasteners **410**. The plurality of second building panels **200** may be secured to the support member **50** by the plurality of first mechanical fasteners **410**.

(146) In securing the second building panel **200** to the support member **50**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that the head **411** of the first fastener **410** is located inset of the first major surface **201** of the second building

panel **200**. State otherwise, in securing the second building panel **200** to one or more of the first building panel **100**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that the head **411** of the first fastener **410** is located inset of the first major surface **201** of the second building panel **200**.

(147) At least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that the head **411** is entirely positioned between the first major surface **201** and the second major surface **202** of the second building panel **200**. At least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that the head **411** of the first fastener **410** does not overlap with the second major surface **201** of the second building panel **200**. At least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that the proximal end of the head **411** is inset from the first major surface **201** of the second building panel **200** by the first depth D.sub.1. The first depth D.sub.1 may range from about 1% to about 30% the second thickness t.sub.2 of the second building panel **200**—including all percentages and sub-ranges there-between.

(148) In securing the second building panel **200** to the support member **50**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that at least a portion of the threaded body **412** of the first fastener **410** is located inset of the first major surface **201** of the second building panel **200**. In securing the second building panel **200** to the support member **50**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that at least a portion of the threaded body **412** of the first fastener **410** is located inset of the first major surface **201** of the second building panel **200** and extends past the first interface **60** and into the support member **50**.

(149) In securing the second building panel **200** to the support member **50**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that at least a first portion of the threaded body **412** of the first fastener **410** is located between the first major surface **201** and the second major surface **202** of the second building panel **200**, and a second portion of the threaded body **412** extends through the first interface **60**, and a third portion of the threaded body **412** extends past the support surface **51** and into the support member **50**.

(150) In securing the second building panel **200** to the support member **50**, at least one of the plurality of first fasteners **410** may be driven into the second building panel **200** such that at least a first portion of the threaded body **412** of the first fastener **410** is located between the first major surface **201** and the second major surface **202** of the second building panel **200**, and a second portion of the threaded body **412** extends through the first interface **60**, and a third portion of the threaded body **412** extends past the support surface **51** and into the support member **50**—whereby the second portion of the threaded body **412** contacts the adhesive **300** present in the first interface **60**.

(151) During installation, the second major surface **202** of at least one of the second building panel **200** is brought into contact with a support surface **51** of the support member **50** to form the first interface **60**, whereby the adhesive **300** is present as the first interface **60**, and the adhesive is in a state where the adhesive **300** has not yet reach full bond strength.

(152) Installation may further comprise driving at least one of the first mechanical fasteners **410** into the at least one second building panel **200** and the support member **50**. The first mechanical fastener **410** may be driven into the second building panel **200** and through the first interface **60** before the adhesive **300** has reached full bond strength. Subsequently, installation may include a time period sufficient for such adhesive **300** to reach full bond strength. After the time period where the adhesive **300** reaches full bond strength, one or more of the plurality of first mechanical fasteners **410** may be removed.

(153) Referring now to FIGS. **8C** and **8D**, the thin dotted linear lines represent the plurality of support members **50** and the thin solid linear lines represent the second building panels **200**, and the thin solid wavy lines represent the adhesive **300**, and the dotted-dashed linear lines represent

the boundary between the central region R.sub.C and the perimeter region R.sub.P of each second building panel **200**. The support members **50** are represented by as thin dotted lines so that it may be shown how the plurality of second building panels **200** are oriented relative to the plurality of support members **50** in forming the facing layer **90** as well as the orientation of the adhesive **300** within the first interface **60** located between the support members **50** and the second building panels **200**.

(154) When installed, the first side surface **203a** of a first one of the plurality of second building panels **200** and the second side surface **203b** of a second one of the plurality of second building panels **200** may both overlap one support member **50**. Adhesive **300** may be present in the first interface **60** formed where the first side surface **203a** of a first one of the plurality of second building panels **200** and the second side surface **203b** of a second one of the plurality of second building panels **200** overlap one support member **50**. A first one of the plurality of first mechanical fasteners **410** may pass through the first one of the plurality of second building panels **200** where that first one of the plurality of second building panels **200** overlaps with the first interface **60** and a second one of the plurality of first mechanical fasteners **410** may pass through the second one of the plurality of second building panels **200** where that second one of the plurality of second building panels **200** overlaps with the first interface **60**.

(155) When installed, the third side surface **203c** of a first one of the plurality of second building panels **200** and the fourth side surface **203d** of a second one of the plurality of second building panels **200** may both overlap one support member **50**. Adhesive **300** may be present in the first interface **60** formed where the third side surface **203c** of a first one of the plurality of second building panels **200** and the fourth side surface **203d** of a second one of the plurality of second building panels **200** overlap one support member **50**. A first one of the plurality of first mechanical fasteners **410** may pass through the first one of the plurality of second building panels **200** where that first one of the plurality of second building panels **200** overlaps with the first interface **60** and a second one of the plurality of first mechanical fasteners **410** may pass through the second one of the plurality of second building panels **200** where that second one of the plurality of second building panels **200** overlaps with the first interface **60**.

(156) When installed, at least one of the plurality of second building panels **200** may be positioned such that the first elongated edge **203a** (first side surface **203a**) and/or the second elongated edge **203b** (second side surface **203b**) overlaps with an elongated one 53 of the support member **50**.

When installed, at least one of the plurality of second building panels **200** may be positioned such that the third side surface **203c** (first transverse edge) and/or the second side surface **203d** (second transverse edge) overlaps with a transverse one 54 of the support member **50**. If during installation a transverse one 54 of the support member **50** does not previously exist, such transverse one 54 of the support member **50** may be installed such that the second building panel **200** can overlap along the third and/or fourth side surface **203c**, **203d**.

(157) As demonstrated by FIGS. **8C** and **8D**, the adhesive **300** may be applied in a plurality of lateral strips. Although not shown, the adhesive **300** may be applied in a plurality of longitudinal strips. The orientation of the strips may be applied such that the adhesive **300** is present in the first interface **60**—regardless of orientation of the second building panel **200** relative to the support members **50**.

(158) As demonstrated by FIG. **8C**, the plurality of second building panels **200** may be secured to the support members **50** by a first plurality of first mechanical fasteners **410a** and second plurality of first mechanical fasteners **410a**—whereby the first plurality of first mechanical fasteners **410a** are located within the central region R.sub.C and the second plurality of second mechanical fasteners **410b** are located within the perimeter region R.sub.P.

(159) According to such embodiments, the perimeter region R.sub.P may have a perimeter distance ranging from about 0.5 inches to about 1.0 inches—including all distances and subranges there-between. According to such embodiments, the first plurality of first mechanical fasteners **410a** may

be present in the central region R.sub.C an amount ranging from about 4 to about 12—including all integers and sub-ranges there-between. In some embodiments, the first plurality of first mechanical fasteners **410a** may be present in the central region R.sub.C an amount ranging from about 4 to about 8—including all integers and sub-ranges there-between. According to such embodiments, the first plurality of first mechanical fasteners **410a** may have a longitudinal separation distance D.sub.2 ranging from about 15 inches to about 21 inches—including all distances and sub-ranges there-between. In some embodiments, the first plurality of first mechanical fasteners **410a** may have a longitudinal separation distance D.sub.2 of about 16 inches to about 18 inches.

(160) The first plurality of first mechanical fasteners **410a** may have a longitudinal separation distance D.sub.2 that is equal to about 15% to about 30% the second length L.sub.2 of the second building panel **200**—including all percentages and sub-ranges there-between.

(161) According to such embodiments, the second plurality of first mechanical fasteners **410b** may be present in the perimeter region R.sub.P an amount such that the second plurality of first mechanical fasteners **410b** are separated by longitudinal separation distance D.sub.2 and a lateral separation distance D.sub.3 ranging from about 6 inches to about 15 inches—including all distances and sub-ranges there-between. The second plurality of first mechanical fasteners **410b** may have a longitudinal separation distance D.sub.2 that is equal to about 6% to about 15% the second length L.sub.2 of the second building panel **200**—including all percentages and sub-ranges there-between.

(162) According to such embodiments, where adhesive **300** is present in the second interface **70**, the first plurality of first mechanical fasteners **410a** and the second plurality of first mechanical fasteners **410b** should extend through the adhesive **300**.

(163) The first plurality of first mechanical fastener **410a** may be present in a first fastener density. The first fastener density may be calculated as the total number of the first plurality of first mechanical fasteners **410a** that are located within the first surface area of the first major surface **201** of the second building panel **200**. The second plurality of first mechanical fastener **410b** may be present in a second fastener density. The second fastener density may be calculated as the total number of the second plurality of first mechanical fasteners **410b** that are located within the first surface area of the first major surface **201** of the second building panel **200**.

(164) The first fastener density and the second fastener density may not be equal. The second fastener density may be greater than the first fastener density. The first fastener density may range from about 0.15 fasteners/ft.sup.2 to about 0.5 fasteners/ft.sup.2—including all densities and sub-ranges there-between. The second fastener density may range from about 0.6 fasteners/ft.sup.2 to about 1.3 fasteners/ft.sup.2—including all densities and sub-ranges there-between. A ratio of the second fastener density to the first fastener density may range from about 1.2:1 to about 8.6:1—including all ratios and sub-ranges there-between.

(165) Referring now to FIG. **8D**, an alternative to the embodiment of FIG. **8C** is shown, whereby the plurality of second building panels **200** and may be secured by a plurality of the first mechanical fasteners **410** and a plurality of the third mechanical fasteners **430**. According to this embodiment, the second building panel **200** may be secured such that the plurality of the first mechanical fasteners **410** are located within the central region R.sub.C of the second building panel **200** and a plurality of the third mechanical fasteners **430** along the edge interface **40** of adjacent second building panels **200**.

(166) According to such embodiments, the perimeter region R.sub.P may be free of the first mechanical fastener **410** whereby only the central region R.sub.C comprises the first mechanical fastener **410**. In other embodiments, the perimeter region R.sub.P may comprise one or more of the first mechanical fastener **410**.

(167) For the third mechanical fasteners **430** located along the edge interface **40**, the lateral separation distance D.sub.3 may range from about 8 inches to about 30 inches—including all distances and sub-ranges there-between.

(168) As demonstrated by FIG. 8D, the plurality of second building panels **200** may be secured to the support members **50** by a plurality of first mechanical fasteners **410** and plurality of third mechanical fasteners **430**. In such embodiments, the edge interface **40** may comprise the plurality of third mechanical fasteners **430**.

(169) Subsequent to application of the concealment compound **500**, one or more coats of a paint composition **600** may be applied to the entirety of the facing surface **91** of the facing layer **90** comprising the concealment compound **500**.

(170) Referring now to FIGS. 9-11, subsequent to application of the concealment compound **500**, one or more coats of a paint composition **600** may be applied to the entirety of the facing surface **91** of the facing layer **90** comprising the concealment compound **500**. Once the facing layer **90** is formed after installing the plurality of second building panels **200**, a concealing composition **500** may be applied to a facing surface **91** of the facing layer **90**. Non-limiting examples of concealment compound **500** include spackle.

(171) According to the present invention, it has been discovered that, by securing the second building panels **200** to the support member **50** with at least the adhesive **300** and at least one of the first mechanical fastener **410** at the first depth D.sub.1, not only are fewer of the first mechanical fasteners **410** required to seismically secure such plurality of second building panels **200**—thereby necessitating fewer first mechanical fasteners **410** to conceal on the facing surface **91**, but the concealment process is also simplified as the hole formed into the second building panel **200** by the first mechanical fastener **410** is more easily filled by the concealment compound **500**. Thus, the present invention provides a new and improvement seismically stable building surface that can be installed with better aesthetics and requiring less skilled labor during the concealment process.

(172) Subsequent to application of the concealment compound **500**, one or more coats of a paint composition **600** may be applied to the entirety of the facing surface **91** of the facing layer **90** comprising the concealment compound **500**. Once the facing layer **90** is formed after installing the plurality of second building panels **200**, a concealing composition **500** may be applied to a facing surface **91** of the facing layer **90**. Non-limiting examples of concealment compound **500** include spackle.

Claims

1. A building system comprising: a plurality of support members located in a plenary space of an interior building space, each support member comprising a support surface; a plurality of first building panels located below the plenary space and secured directly to the support surface of the plurality of support members, each of the plurality of first building panels comprising a non-porous body, each of the first building panels comprising a first major surface opposite a second major surface and a side surface extending between the first major surface and the second major surface; and a plurality of second building panels secured directly to the plurality of first building panels by an adhesive and a plurality of first mechanical fasteners, each of the plurality of the second building panels comprising a porous body, each of the second building panels comprising a first major surface opposite a second major surface and a side surface extending between the first major surface and the second major surface; wherein the second major surface of the plurality of first building panels faces the support surface of the plurality of support members; and wherein the second major surface of the plurality of the second building panels faces the first major surface of the plurality of first building panels.

2. The building system according to claim 1, wherein the plurality of first mechanical fasteners each comprise a head and a threaded body, and wherein the threaded body of one of the plurality of first mechanical fasteners extends through at least a portion of the porous body of one of the plurality of second building panels and at least a portion the non-porous body of one of the plurality of first building panels, wherein the head of at least one of the plurality of first mechanical

fasteners is located between the first major surface and the second major surface of the plurality of second building panels.

3. The building system according to claim 1, wherein a first interface exists between the second major surface of the plurality of first building panels and the support surface of the plurality of support members and a second interface exists between the second major surface of the plurality of second building panels and the first major surface of the plurality of first building panels, wherein the adhesive is present in the second interface, and wherein the first interface is substantially free of the adhesive.

4. The building system according to claim 3, wherein the threaded body of the plurality of first mechanical fasteners extends through the second interface, and wherein the threaded body of at least one of the plurality of first mechanical fasteners does not extend through the first interface.

5. The building system according to claim 1, wherein the non-porous body comprises dry-wall, gypsum board, plywood, particle board, and combinations thereof, wherein the porous body is formed of a fibrous material, wherein the fibrous material is an inorganic fiber, and wherein the inorganic fiber is selected from the group consisting of mineral wool, rock wool, stone wool, and glass fibers.

6. The building system according to claim 1, wherein the non-porous body has a first body density and the porous body has a second body density, wherein the first body density and the second body density are not equal, wherein the first body density ranges from about 16 lb./ft.³ to about 180 lb./ft.³, wherein the second body density ranges from about 2 lb./ft.³ to about 16 lb./ft.³, and wherein a ratio of the first body density to second body density ranges from about 1.5:1 to about 10:1.

7. The building system according to claim 1, further comprising a plurality of second mechanical fasteners that each comprise a head and a threaded body, and wherein the threaded body of the plurality of second mechanical fasteners extends through at least a portion of the non-porous body of one of the plurality of first building panels and the support surface of one of the plurality of support members, wherein the side surface of the first building panel defines a perimeter of the first building panel, and wherein the plurality of first mechanical fasteners are present within the perimeter of one of the second building panels in a first fastener density and the plurality of second mechanical fasteners are present within the perimeter of one of the first building panels in a second fastener density, wherein the first fastener density and the second fastener density are not equal.

8. The building system according to claim 7, wherein the second fastener density is greater than the first fastener density, wherein the first fastener density ranges from about 0.2 fasteners/ft.² to about 0.75 fasteners/ft.², wherein a ratio of the second fastener density to first fastener density ranges from about 1.1:1 to about 10:1.

9. The building system according to claim 1, wherein each of the plurality of first building panels has a first thickness as measured between the first major surface and the second major surface of the first building panels, and each of the plurality of second building panels has a second thickness as measured between the first major surface and the second major surface of the second building panels, and wherein the first thickness and the second thickness are not equal.

10. The building system according to claim 9, wherein the first thickness ranges from about 0.3 inches to about 0.75 inches, wherein the second thickness ranges from about 0.4 inches to about 1.6 inches, wherein a ratio of the second thickness to the first thickness ranges from about 3.0:1 to about 1.25:1, wherein each of the plurality of first mechanical fasteners has a first fastener length, and wherein a ratio of first fastener length to second thickness ranges from about 1.25:1 to about 2.5:1.
